

Analog System Design Lab Mini-Project Report

ANALOG SOIL MOISTURE CONTROLLER

Submitted by

Roll NO.	Registration No.	Student Name	Section
4	240906022	HRISHIKESH VINAYAK DHABADE	C1
6	240906052	AKSHITH G RAO	C1
15	240906284	AKRATI SINGH	C1
18	240906416	AAYANSH TRIPATHI	C1



SCHOOL OF ELECTRICAL ENGINEERING
MANIPAL INSTITUTE OF TECHNOLOGY
(A Constituent Unit of Manipal Academy of Higher Education)

TABLE OF CONTENTS

Acknowledgement	4
Abstract	5

Chapter 1	INTRODUCTION	6
1.1	Introduction	6
1.2	Motivation	6

Chapter 2	BACKGROUND THEORY AND OBJECTIVES	7
2.1	Background Theory	7
2.2	Objective	7

Chapter 3	METHODOLOGY	8
3.1	Block diagram	8
3.2	Specification	8-9
3.3	Design and implementation	10

Chapter 4	RESULT	12
4.1	Final circuit diagram and result	12

Chapter 5	CONCLUSION AND FUTURE WORK	13
5.1	Summary	13
5.2	Conclusion	13
5.3	Future Scope	13

REFERENCE
APPENDIX
INDIVIDUAL CONTRIBUTION
PLAGIARISM AND AI REPORT

ACKNOWLEDGMENT

We would like to express our sincere gratitude to everyone who supported us throughout this mini project.

First and foremost, we thank our course instructor and lab guide for their constant guidance and encouragement during the project. Their insights on analog circuit design and timely feedback helped us stay on the right track and resolve the challenges we faced during both simulation and hardware testing.

We are also grateful to the School of Electrical Engineering, Manipal Institute of Technology, for providing us with the necessary lab facilities, components, and LTspice software access that made this project possible.

We would like to thank our parents and families for their continuous support and motivation throughout our academic journey.

Finally, we thank our teammates for the effort, cooperation, and dedication each one of us put into this project. Every member contributed meaningfully — from circuit design and simulation to hardware testing and report writing — and we are proud of what we accomplished together as a team.

Hrishikesh Vinayak Dhabade

Akshith G Rao

Akrati Singh

Aayansh Tripathi

C1

Electrical and Electronics engineering

School of Electrical Engineering Manipal Institute of Technology

ABSTRACT

Irrigation management remains one of the biggest challenges in modern agriculture, especially in areas where water availability is low and constant manual monitoring is not possible. This project aims to solve this problem by designing an analog closed-loop soil moisture controller that can automatically control irrigation systems without the use of microcontrollers or any software.

The basic idea behind this project is to exploit the fact that the resistance to dry soil increases. For this, two probes are inserted into the soil, which generate a voltage based on the moisture content. This voltage signal is then given to a NE555 Timer IC, which works in Schmitt Trigger mode. If the soil becomes too dry, the voltage becomes too low (one-third of the total voltage), and the IC output becomes HIGH.

This HIGH output then goes to a BC547 transistor, which then switches ON a relay connected to a DC water pump. The water then starts flowing into the soil. A potentiometer (15k-100k ohm) can be used to adjust the moisture content according to the crop type.

When water is added to the soil and it becomes moist, the resistance decreases, causing the voltage to rise to a HIGH value, which then becomes LOW, switching off the pump. This design can be used to control irrigation systems automatically, and the results were verified by simulating the design in LTspice and then building it on a breadboard.

The results showed that the design can be used to automatically switch the pump ON/OFF based on the moisture content in the soil. This design can be used to control irrigation systems automatically, and the results were verified by simulating the design in LTspice and then building it on a breadboard. This design can be used to control irrigation systems automatically, and the results were verified by simulating the design in LTspice and then building it on a breadboard.

INTRODUCTION

Water is one of the most important resources used in agriculture, but still, every day, a huge amount of water is wasted due to inefficient irrigation methods. In the traditional system of farming, the irrigation system is controlled by the farmer's observations. The farmer observes the soil condition, decides whether the soil requires irrigation, and then irrigates the fields accordingly. This is not only an inefficient system but also often proves to be detrimental to the growth of the crops. Sometimes the crops are irrigated with too much water, while at other times the crops are not provided with the required amount of water, which is not suitable for the growth of the crops. Therefore, there is an obvious need for an efficient irrigation system.

The motivation behind this project is the above-mentioned inefficient irrigation system. There are many efficient irrigation systems available nowadays, but almost all the systems make use of microcontrollers. Even though the systems make use of microcontrollers, which is an efficient way, the complexity involved in the system is unnecessary.

In order to resolve this, an analog closed-loop soil moisture controller is implemented using a Timer IC, specifically the NE555 Timer IC, completely avoiding digital components. The NE555 IC, when connected to work in Schmitt Trigger mode, acts as a comparator to compare the voltage generated by the soil probes. This voltage will decrease as the soil gets drier, since the soil's resistance increases when it gets drier. This voltage will be compared to a set threshold voltage, and when it goes below a certain level, a relay will be switched on to power a DC water pump. This whole process will be done automatically.

This project not only resolves a real-world agricultural problem, but it also teaches a lot about analog electronic circuit design. This project will be helpful in understanding some basic analog electronic circuit design concepts, such as voltage division, transistor switching etc. All these concepts are basic concepts in electronic circuit design, and they will be very helpful to anyone who wants to work in electronic circuit design.

BACKGROUND THEORY AND OBJECTIVE

BACKGROUND THEORY:

Soil moisture monitoring is an important aspect that has been researched extensively in agricultural engineering as well as electronics. The majority of the low-cost soil moisture sensing devices work on the basic principle that the electrical resistance between two points varies depending on the moisture content in the soil. When the soil is damp, the electrical conductivity is high due to the ions present in the water. As the soil dries out, the conductivity reduces, resulting in an increased resistance between the two points.

NE555 Timer IC is one such IC used in the project, which is an analog IC used for various purposes. The IC is used in the Schmitt Trigger mode, where the IC acts as a comparator with hysteresis. This means that the IC will operate with two thresholds: one at one-third voltage supply and another at two-thirds voltage supply. This is an important aspect as the IC will not switch rapidly between the on and off states when the voltage supply is close to the threshold voltage. This is also known as chattering. The NE555 IC is used in various sensor-based circuits as the IC is easy to work with, especially when the circuit does not call for the use of a microcontroller.

In order to switch the pump on and off, the NE555 IC is connected to an NPN transistor. The transistor is used as a current amplifier as the relay requires more current than the IC is capable of supplying.

OBJECTIVES:

The main objectives of this project are as follows:

To design a voltage divider circuit using soil probes and a potentiometer that can produce a suitable input voltage for the NE555 circuit.

To design a transistor and relay-based circuit that can operate a DC water pump.

To test and verify the complete circuit system by ensuring that the pump is turned on automatically when the soil is dry and turned off when there is enough moisture in the soil using LTspice simulation and hardware implementation on a breadboard.

DESIGN AND METHODOLOGY

1. Specifications

The analog soil moisture controller was designed to meet the following system specifications:

Supply Voltage: 9V–12V DC

Sensing Method: Resistive soil probes

Control IC: NE555 Timer in Schmitt Trigger (Bistable) mode

Switching Device: BC547 NPN transistor driving a 9V/12V SPDT relay

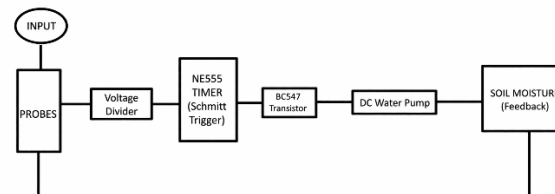
Moisture Threshold: Adjustable via 15k–100k ohm potentiometer

Output Load: Small DC water pump

Indicators: Piezo buzzer for pump activation feedback

Platform: Breadboard prototype validated by LTspice simulation

2. Block Diagram



3. Design of each block, steps with calculations

a. Soil Probe and Voltage Divider:

The soil probes are two metal electrodes inserted into the soil. When soil is wet, it conducts well and resistance between the probes is low. As it dries out, resistance increases. We used this property by connecting the probes in series with a 100k ohm potentiometer to form a voltage divider, whose output goes into the NE555 timer.

When soil is moist, the midpoint voltage stays high. When dry, it drops. The potentiometer lets us adjust the sensitivity — setting the exact moisture level at which the pump should activate.

Using the voltage divider formula:

$$V_{in} = V_{cc} \times R_{pot} / (R_{soil} + R_{pot})$$

For $V_{cc} = 9V$, the NE555 lower threshold is $V_{cc}/3 = 3V$. With R_{pot} set to 33k ohm, the pump triggers when R_{soil} exceeds approximately 99k ohm, which corresponds to dry soil conditions.

b. NE555 Timer in Schmitt Trigger Mode:

We chose the NE555 because it works as a comparator with hysteresis. In Schmitt Trigger mode, pins 2 and 6 are connected together and fed the voltage divider output. The IC compares this against two internal thresholds — $V_{cc}/3$ and $2V_{cc}/3$:

Input drops below $V_{cc}/3$ (3V) → output goes HIGH → pump ON

Input rises above $2V_{cc}/3$ (6V) → output goes LOW → pump OFF

The gap between these thresholds prevents the pump from repeatedly switching ON and OFF when moisture is near the boundary level. A 10pF capacitor on pin 5 filters noise on the internal reference, keeping the switching stable.

c. Transistor Switching Stage:

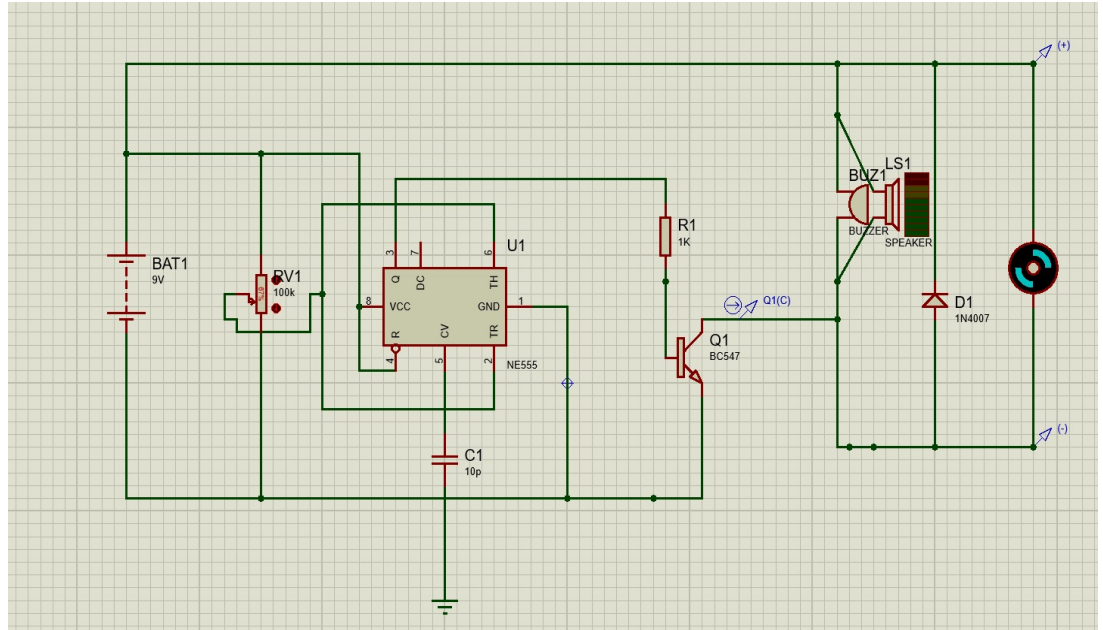
We used a BC547 NPN transistor to drive the relay coil without stressing the NE555. When the output goes HIGH, current flows through the 1k ohm base resistor into the transistor base, saturating it and allowing current to flow through the relay coil.

Base current works out to:

$$I_B = (9 - 0.7) / 1000 \approx 8.3 \text{ mA}$$

With BC547 gain (hFE) of around 110, the transistor can comfortably handle the relay coil current of 50–100mA, ensuring clean and reliable switching.

4. Circuit diagram – Each block circuit diagram and complete integration circuit schematic



- a. Power Supply: BAT1, a 9V battery, is used as a power source for the circuit. The positive terminal is connected to Vcc and the negative terminal is connected to common ground.
- b. Sensing Stage: RV1 is a 100K potentiometer connected in series across Vcc and the input pins of the NE555 integrated circuit. It is connected in parallel with the soil probes, denoted by the (+) and (-) terminals on the extreme right side of the circuit diagram. The wiper terminal is connected in parallel across both pin 2 and 6 of the NE555.
- c. NE555 Timer: The NE555 integrated circuit is denoted by U1 in the circuit diagram. It is connected as a Schmitt trigger circuit. Pin 8 is connected to Vcc, pin 1 is connected to ground, pin 2 and 6 are connected in parallel across RV1, pin 4 is connected to Vcc as a trigger circuit, and pin 5 is bypassed to ground through a 10pF capacitor denoted by C1. Pin 3 is connected in series across a 1K resistor denoted by R1 and transistor Q1.
- d. Switching and Output Stage: Q1 is a BC547 transistor connected in series across a buzzer denoted by BUZ1 and a speaker denoted by LS1. These two devices are switched on when Q1 is switched on by a high signal from pin 3 of the NE555 integrated circuit. D1 is a 1N4007 diode connected in reverse bias across the load circuit and is used as a back-EMF protector for transistor Q1. The DC pump is denoted by a circle on the extreme right side of the circuit diagram and is connected in series across the positive and negative rails.

5. Explanation of Functionality

The way in which the circuit operates is actually quite simple once you know what each part of the circuit is doing. As soon as you turn on the circuit, the soil probes will continue to monitor the soil resistance. If the soil is moist, then the resistance is low, and therefore the voltage divider output is high, i.e., above $2V_{cc}/3$. This ensures that the NE555 is in a LOW state, keeping the transistor switched OFF.

As the soil dries up, the resistance increases, causing the divider voltage to decrease. As soon as the divider voltage drops below $V_{cc}/3$, the NE555 output goes HIGH. This causes the BC547 transistor to switch on fully, which in turn switches on the relay and activates the pump.

As soon as the pump injects water into the soil, the resistance decreases once again. As soon as the divider voltage is once again above $2V_{cc}/3$, the NE555 switches LOW, and the transistor switches OFF, causing the pump to stop. This cycle repeats itself automatically without any intervention from us, which is exactly what we were hoping for in a closed-loop system.

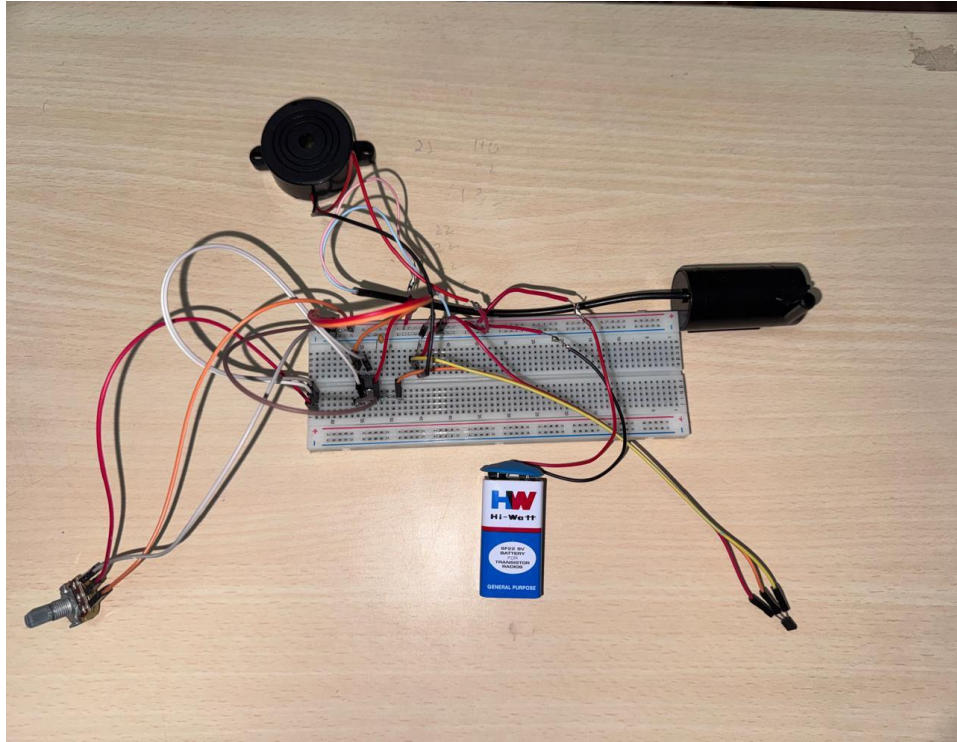
6. Application

This project has a number of practical applications. The most obvious one is in home gardening. A person might forget to water their plants regularly; however, this circuit takes care of it automatically. It can also be used in a small greenhouse where soil moisture is a crucial factor in maintaining quality crops.

Apart from these applications, this circuit can be used as a demonstration circuit in educational institutions. It is an effective way of demonstrating how analog components like the NE555, transistor, and relay can be used to create a working circuit without a microcontroller. It can also be used in remote areas where power consumption is a low-priority factor.

RESULT

Final circuit diagram (with all hardware):



The designed analog soil moisture controller circuit was successfully implemented on the breadboard. The circuit performed as required when the soil moisture level varied. When the soil condition was such that the moisture level was low, the voltage from the sensing stage was reduced below the threshold voltage, and the NE555 timer output was set to HIGH. This made the transistor, relay, buzzer, and pump turn ON.

When the soil condition was improved by the addition of moisture, the voltage from the sensing stage was increased, and the output from the NE555 timer was set LOW, turning the pump OFF.

The practical implementation of the circuit showed that the system was working well with the required switching action at the desired soil condition. The circuit was also able to adjust the threshold voltage with the help of the potentiometer, depending on the condition of the soil. The final hardware implementation is shown.

CONCLUSION AND FUTURE WORK

CONCLUSION:

This project has successfully shown the viability of implementing a fully functional soil moisture controller using only analog circuits, without the aid of any sort of microcontroller or programming. The NE555 timer chip in its Schmitt Trigger configuration has been seen to be useful in this application. The closed loop behaviour, where the pump is automatically switched on when the soil is dry and automatically switched off when the moisture is replenished, worked exactly as expected.

This circuit is simple to design and implement, yet it is cost-effective and rugged in its operation. It also reinforced our understanding of several important concepts, including voltage dividers, BJT switching circuits, relay driver circuits, hysteresis, and closed loop feedback, which are all important in the design of any analog circuit. It is our belief that this project is extremely useful and relevant to the real world.

FUTURE WORK:

Although the existing solution is good for basic irrigation control needs, there are a number of changes that we wish to implement in the future.

- Multiple soil zones: The existing solution only supports a single probe location. The new version could have multiple probe pairs and use multiple NE555 stages for independent control of irrigation for different zones or plant beds.
- Solar powered operation: The solution requires very little power. Therefore, it could be made solar powered by adding a small solar panel and a backup power source. This will make it a fully sustainable solution.
- Adjustable time delay: The existing solution stops the pump operation as soon as the moisture is detected. A monostable stage could be added to maintain the pump operation for a fixed period of time after the detection of moisture. This will be more practical for a large volume of soil.
- Moisture level display: A simple analog display could be added to visually indicate the moisture level. This will make the solution even more user friendly.

REFERENCES

1. LTspice, "LTspice XVII Simulation Tool," Analog Devices Inc. [Online]. Available: <https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html>
2. ElectroDuino, "Soil Moisture Sensor — How It Works," [Online]. Available: <https://www.electroduino.com/soil-moisture-sensor>. [Accessed: Mar. 2025].
3. Vishay, "1N4007 General Purpose Rectifier Datasheet," Vishay Intertechnology, 2009. [Online]. Available: <https://www.vishay.com/docs/88369/1n4007.pdf>
4. Fairchild Semiconductor, "BC547 NPN Transistor Datasheet," 2002. [Online]. Available: <https://www.onsemi.com/pdf/datasheet/bc547-d.pdf>
5. Texas Instruments, "NE555 Precision Timer Datasheet," Texas Instruments Inc., 2014. [Online]. Available: <https://www.ti.com/lit/ds/symlink/ne555.pdf>

APPENDIX

1. NE555 Timer IC:
 - Operating voltage: 4.5–16 V
 - Threshold voltage: $2V_{cc}/3$
 - Trigger voltage: $V_{cc}/3$
 - Output current: up to 200 mA

2. BC547 Transistor:
 - Type: NPN BJT
 - Collector current: 100 mA
 - Voltage rating: 45 V
 - Gain (h_{FE}): 110–800

3. 1N4007 Diode:
 - Maximum reverse voltage: 1000 V
 - Forward current: 1 A

4. Relay:
 - Type: 9V SPDT
 - Used for pump switching

5. Potentiometer:
 - Value: 100 k Ω
 - Used for moisture threshold adjustment

INDIVIDUAL CONTRIBUTION

Hrishikesh Vinayak Dhabade:

Hrishikesh contributed to designing the circuit for the project. He also helped in selecting the required components for the circuit. Moreover, he contributed to the simulation of the circuit by using LT spice. Along with these responsibilities, he helped in the preparation of the final report.

Akshith G Rao:

Akshith helped in the construction of the circuit on the breadboard. Moreover, he helped in checking the working of the circuit for different soil conditions. Along with these responsibilities, he contributed in the writing of the report.

Akrati Singh:

Akrati contributed in the research and documentation of the project. She helped in designing the block diagram of the circuit. Moreover, she explained the working principle of the circuit. Along with these responsibilities, she contributed in the writing of the report.

Aayansh Tripathi:

Aayansh contributed in the simulation and practical implementation of the circuit. Moreover, he helped in testing the circuit. Along with these responsibilities, he contributed in the writing of the report.

PLAGIARISM REPORT



1. <https://www.electricaltechnology.org/2020/04/npn-transistor.html>
2. <https://www.electronicshub.org/transistor-as-a-switch/>
3. <https://testbook.com/electrical-engineering/transistor-as-an-amplifier-principles-and-working>

AI REPORT

AI Detector - Trusted AI Check

Ask Gemini

zerogpt.com

Experience the best with our premium plans – unlock exclusive features now!

ZeroGPT

en

Get Started

Register

Login



this circuit takes care of it automatically. It can also be used in a small greenhouse where soil moisture is a crucial factor in maintaining quality crops.

Apart from these applications, this circuit can be used as a demonstration circuit in educational institutions. It is an effective way of demonstrating how analog components like the NE555, transistor, and relay can be used to create a working circuit without a microcontroller. It can also be used in remote areas where power consumption is a low-priority factor.

Detect Text

Upload File

14,999/15,000 Characters
Check 350,000 characters, [Upgrade Here](#)

Your File Content is Human written



7.3%
AI GPT*